In Class Template

Ethan C

1. Set Up

Packages

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(here)
```

```
here() starts at /Users/ethancastelazo/GitHub/week-05_spring-2026_aquatic-
inverts
```

```r
library(janitor)
```

```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(snakecase)

#Wrap Axis Label
library(scales)
```

```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
```

Data

```r
# taxnomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
Rows: 50 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)
```

2. Cleaning

`sites` object

Goal: createa a new object from `sites` that only includes NCOS sites

```
# creating new object from sites
ncos_sites <- sites %>%
  # filering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

clean taxon list

Goal: convert character strings in the existing `snakecase` data frame into snakecase

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean `water_quality`

Goal: create a data frame with median pH, median DO (mg/L), mean salinity (ppt) for each date and site

- create a new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames = group by date_site
- calculate median pH, salinity, and DO
- ungroup
- filter out any outlines

```
water_quality_clean <- water_quality %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  unite("date_site", date, site, remove = TRUE) %>%
  group_by(date_site) %>%
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
  ungroup() %>%
  filter(date_site != "2022-08-12NVBR")
```

clean `aq_ins`

Goal: create a long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality information for each survey.

- create a new object from `aq_ins`
- clean column names
- filter to only include quarterly NCOS sites
- select columns
- filer to only include "filtered beaker" sample
- convert the data frame to long format
- group by site, date, and taxon
- calculate average abundance per liter
- create a new column called `date_site` to join with water quality
- join data frames
- join taxon list

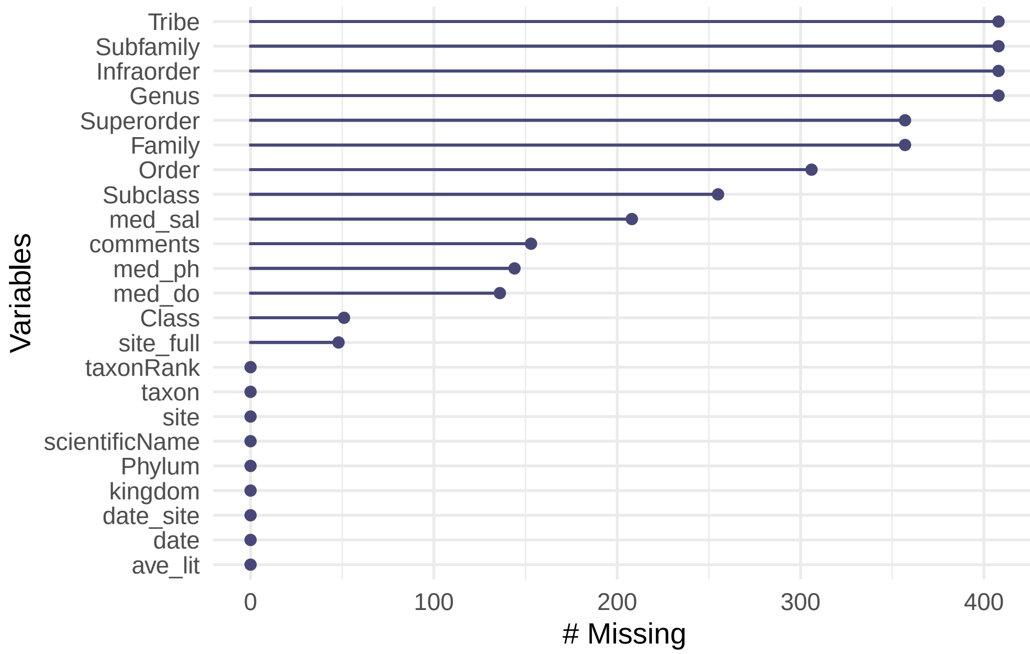
```r
aq_ins_clean <- aq_ins %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  rename(date = date_on_vial) %>%
  select(date, sample_type,site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  ungroup() %>%
  unite("date_site", date, site, remove = FALSE) %>%
  left_join(water_quality_clean, by = "date_site") %>%
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "West Pond",
    site == "NDV" ~ "Devereux Creek")) %>%
  mutate(site_full = fct_reorder(site_full,
                                 med_sal,
                                 .fun = "median",
                                 .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

```
`summarise()` has grouped output by 'date', 'site'. You can override using the
`.groups` argument.
```

3. Visualizing

Visualize missing values

```r
gg_miss_var(aq_ins_clean)
```

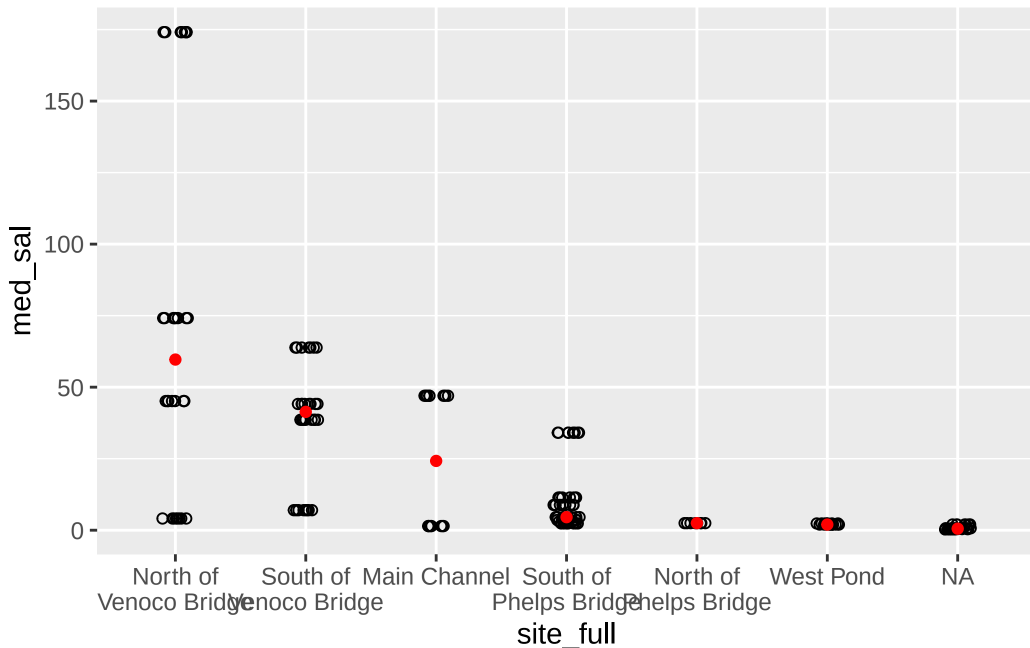



```r
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21)+
  stat_summary(geom = "point",
               fun = median,
               color = 'red')+
  scale_x_discrete(labels = label_wrap(width = 15))
```

```r
salinity_plot
```

```
Warning: Removed 208 rows containing non-finite outside the scale range
(`stat_summary()`).
```

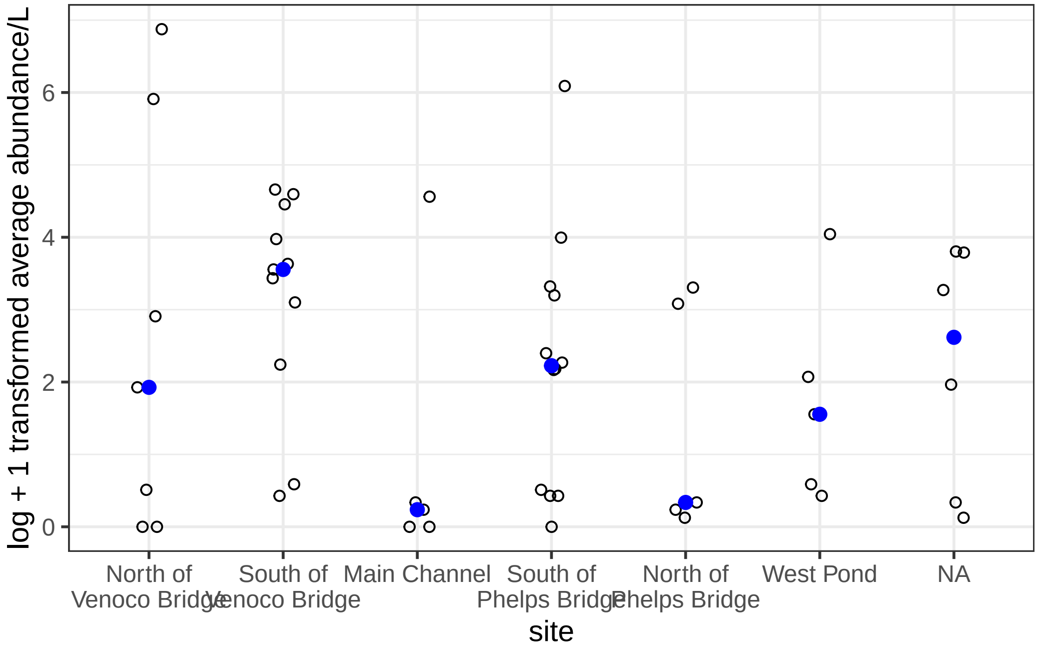
```
Warning: Removed 208 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
copepods <- aq_ins_clean %>%
  filter(taxon == "copepod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))
```

```r
copepod_plot <- ggplot(data = copepods,
                       mapping = aes(x = site_full,
                                     y = log_ave_lit))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
```

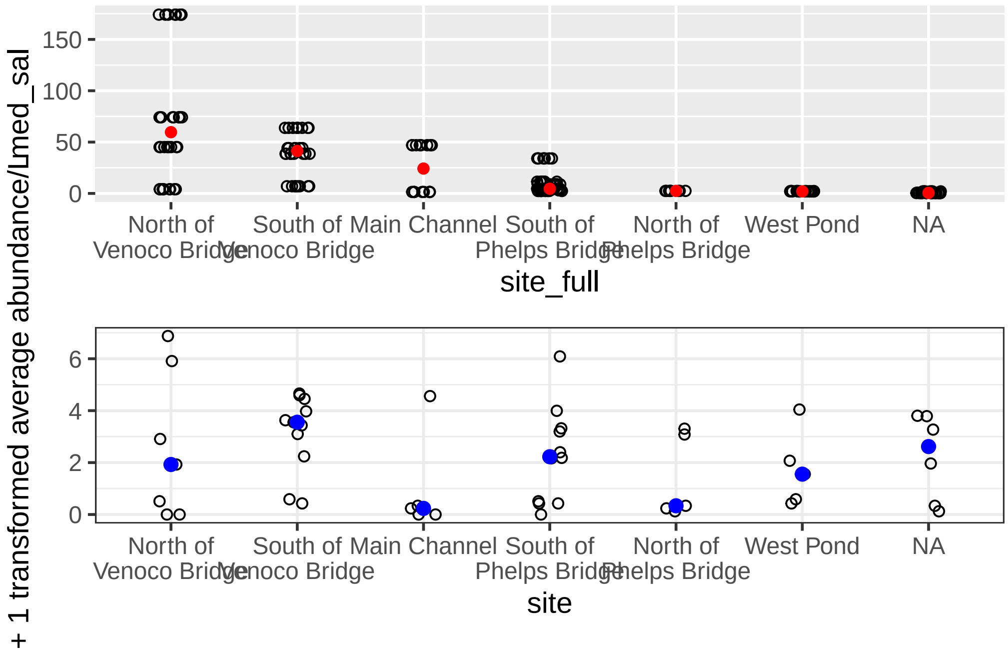
```
                color = "blue",
                size = 2)+
  scale_x_discrete(labels = label_wrap(width = 15)) +
                       labs(x = "site",
                            y = "log + 1 transformed average abundance/L") +
                       theme_bw()
copepod_plot
```




```
salinity_plot / copepod_plot
```

```
Warning: Removed 208 rows containing non-finite outside the scale range
(`stat_summary()`).
```

```
Warning: Removed 208 rows containing missing values or values outside the scale range
(`geom_point()`).
```


+ 1 transformed average abundance/Lmed_sal
150
100
50
0
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
West Pond
NA
site_full
6
4
2
0
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
West Pond
NA
site